

FJFI - Department of Physics
02MECH - Mechanics
Makeup

01.03.2024
14:00

Name					
Problem	1	2	3	4	Total
Score					

Instructions

1. The time allocated for the exam is 120 minutes.
2. During the exam, you are not allowed to use any class materials or ask your neighbor for help.
3. Calculators are not allowed. You may only have writing supplies and a transparent water bottle without labels on your desk.
4. Use only pens or non-erasable ink pens for the test; erasable pens are not allowed.
5. Write your answers legibly. Illegible answers will not be scored.
6. The maximum score for the test is 4 points.
7. Each question is worth 1 point, but the questions may not require the same amount of time. Therefore, it is advisable to read the entire assignment before starting to solve it.
8. When grading the test, only the solutions provided on the distributed sheets will be considered.
9. Always provide your solutions in separate sheets.

Question 1

A projectile with a mass m strikes a wall with a velocity v_1 and exits from the other side with a velocity v_2 after penetrating the wall of thickness d . What was the average force F acting on the projectile inside the wall?

Solution:

Question 2

Calculate the acceleration of a uniform *cylinder* as it rolls without sliding in a rough inclined plane inclined at an angle α to the horizontal, in a uniform gravitational field of magnitude g .

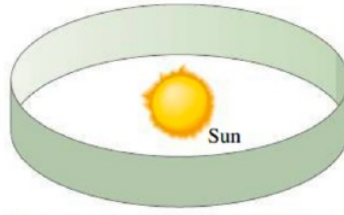
Solution:

Question 3

Determine the moment of inertia for a thin non-homogeneous rod of length l . The linear mass density of the rod shows a linear decrease from λ_1 to λ_2 along its length. Calculate the moment of inertia concerning an axis passing through the rod's center and perpendicular to its length.

Solution:

Question 4



A science-fiction story describes an artificial “planet” in the form of a band completely encircling a sun, as in the figure. The inhabitants live on the interior surface (where it is always noon). Imagine that this sun is exactly like our own, that the distance to the band is the same as the Earth-Sun distance and that the ring rotates sufficiently quickly to produce an apparent gravity of g , as on Earth. What will be the period of revolution, i.e., this planet’s year, in terms of the following quantities?

- g : Gravitational acceleration on the Earth
- r : Distance from the sun to the ring-planet
- M : Mass of the sun
- G : Universal gravitational constant
- Gravitational force between masses M and m : $F = G \frac{Mm}{r^2}$

Solution: